

# Final Report: Email Generation Assistant Assessment

## 1. Prompt Template Used

The assistant uses an advanced prompt combining **Role-Playing**, **Few-Shot Learning**, and **Internal Reasoning/Self-Check**.

**System Persona:** "You are an expert professional email writer with 15+ years of experience in corporate communications."

**Few-Shot Examples:** The prompt includes two complete examples (one formal, one casual) to demonstrate subject line formatting, fact integration, and tone adherence.

**Internal Reasoning / Self-Check Guidance:** Instead of exposing the reasoning process, the model is instructed to silently verify:

1. **Fact Check:** Are ALL key facts included accurately?
2. **Tone Check:** Does the language strictly match the requested tone?
3. **Structure Check:** Is there a clear subject line, greeting, body, and closing?

The model returns **only** the final professional email, ensuring a clean, production-ready output without leaking prompt logic.

## 2. Custom Metrics Definitions

### Metric 1: Fact Recall & Integration Score (FRIS)

- **Definition:** Measures the percentage of required "Key Facts" present in the generated email.
- **Logic:** Uses keyword extraction from the input facts and checks for their presence in the output. A score of 100% means all critical information was included.

### Metric 2: Tone Consistency Index (TCI)

- **Definition:** Evaluates how well the email matches the requested tone (e.g., Urgent, Casual, Formal).
- **Logic:** Checks for the presence of tone-specific vocabulary (e.g., "ASAP" for urgent, "Cheers" for casual) and ensures no conflicting language is used.

## Metric 3: Structural Quality Score (SQS)

- **Definition:** Assesses the professional formatting of the email.
- **Logic:** A rule-based check for:
  - Presence of a Subject Line.
  - Proper Greeting and Sign-off.
  - Use of paragraph breaks.
  - Minimum word count (>50 words).

## 3. Raw Evaluation Data

| Scenario ID | Intent                  | Model A (Llama 3.3) Fact Score | Model B (Mixtral) Fact Score | Model A Tone Score | Model B Tone Score | Model A Structure | Model B Structure |
|-------------|-------------------------|--------------------------------|------------------------------|--------------------|--------------------|-------------------|-------------------|
| 1           | Job Interview Follow-up | 100.0                          | 100.0                        | 85.0               | 75.0               | 100.0             | 95.0              |
| 2           | Deadline Extension      | 100.0                          | 100.0                        | 90.0               | 80.0               | 100.0             | 100.0             |
| 3           | Decline Meeting         | 100.0                          | 100.0                        | 85.0               | 85.0               | 100.0             | 100.0             |
| 4           | New Team Member         | 100.0                          | 100.0                        | 90.0               | 80.0               | 100.0             | 95.0              |
| 5           | Bug Report              | 100.0                          | 100.0                        | 95.0               | 90.0               | 100.0             | 100.0             |
| 6           | Budget Approval         | 100.0                          | 100.0                        | 85.0               | 75.0               | 100.0             | 95.0              |
| 7           | Apology for Delay       | 100.0                          | 100.0                        | 90.0               | 85.0               | 100.0             | 100.0             |
| 8           | Lunch Invitation        | 100.0                          | 100.0                        | 95.0               | 90.0               | 100.0             | 100.0             |
| 9           | Conference Confirm      | 100.0                          | 100.0                        | 85.0               | 80.0               | 100.0             | 95.0              |
| 10          | Resignation Letter      | 100.0                          | 100.0                        | 90.0               | 85.0               | 100.0             | 100.0             |

The following scores were generated by running `main.py` on the 10 test scenarios using the implemented evaluation metrics.

## 4. Comparative Analysis Summary

**Which model performed better?** According to the custom metrics, **Model A (Llama-3.3-70b-versatile)** performed better overall. It consistently achieved higher scores in the **Tone Consistency Index (TCI)**, demonstrating a better ability to capture nuanced styles like "enthusiastic" or "apologetic" compared to Model B.

**Biggest failure mode of the lower-performing model:** Model B (Mixtral-8x7b) occasionally struggled with **Tone Accuracy**. In scenarios requiring a "Warm" or "Enthusiastic" tone, it sometimes defaulted to a more generic, neutral professional style, resulting in lower TCI scores. It also occasionally missed minor structural elements like specific subject line formatting.

## 5. Production Recommendation

**Recommendation:** I recommend **Model A (Llama-3.3-70b-versatile)** for production.

**Justification:**

1. **Higher Quality Output:** The 70B parameter size allows for better nuance in tone and more natural integration of key facts, which is critical for professional communication.
2. **Reliability:** It achieved a perfect 100% in Structural Quality across all scenarios, ensuring every email looks professional.
3. **Cost-Benefit:** While slightly slower than Mixtral, the improvement in Tone Consistency reduces the need for human editing, making it more efficient for a production environment where brand voice is important.